

# IKJOT SIDHU

Bioinformatician | Single-cell and spatial transcriptomics | Jersey City, NJ | (857) 316-5448 | ikjot1000@gmail.com |  
LinkedIn | GitHub | ORCID

## SUMMARY

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Bioinformatician with more than seven years of experience analyzing single-cell RNA-seq, spatial transcriptomics, bulk RNA-seq, and chromatin data. Develops reproducible analysis pipelines and applies statistical and machine-learning methods to biological datasets. Research experience includes skin inflammation, tissue repair, microbiome analysis, genetic association data, and animal behavior. Co-first author of a *Science Immunology* paper on spatial transcriptomics in psoriasis.

## TECHNICAL SKILLS

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**Programming:** R, Python, Bash, SQL

**Single-cell and spatial:** scRNA-seq, single-cell multiome RNA + ATAC, CITE-seq, Visium, Visium HD, Seurat, Scanpy, Cell Ranger, Space Ranger, RCTD

**Sequencing analysis:** bulk RNA-seq, ATAC-seq, ChIP-seq, 16S rRNA sequencing, differential expression, pathway analysis, cell-type annotation

**Genetics and microbiome:** GWAS Catalog, UK Biobank, SNP annotation and mapping, QIIME 2, downstream microbiome analysis in R

**Machine learning:** Applied classification, regression, clustering, dimensionality reduction, and deep learning; training and fine-tuning models for multi-animal pose tracking

**Workflows and computing:** Snakemake, Git, Conda, Docker, LSF, SLURM, high-performance computing

## EXPERIENCE

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### Bioinformatician II | Icahn School of Medicine at Mount Sinai

*Naik Lab, Department of Immunology and Immunotherapy | New York, NY | July 2024 - Present*

- Lead analysis of sequencing studies focused on skin inflammation, tissue repair, and immune-cell function.
- Developed and maintain a reusable pipeline for Visium HD data at 8 micrometer and 16 micrometer bin sizes.
- Analyze and integrate single-cell RNA-seq, spatial transcriptomics, and ATAC-seq data from human and mouse studies.
- Analyze single-cell multiome RNA and ATAC data and map variants from GWAS Catalog and UK Biobank datasets to cell-type-specific accessible chromatin regions.
- Train and fine-tune a deep-learning model for multi-animal pose tracking and analyze movement and behavioral measurements from nearly 100 video recordings.
- Maintain organized datasets, analysis code, and metadata for ongoing laboratory projects.
- Mentor two junior bioinformatics analysts in sequencing analysis, reproducible coding practices, and data visualization.
- Work with experimental scientists and postdoctoral researchers to plan analyses, interpret results, and prepare figures and manuscripts.
- Assist the principal investigator with peer review of manuscripts, focusing on analysis methods, results, and presentation.

### Senior Bioinformatics Programmer | NYU Langone Health

*Applied Bioinformatics Laboratories - Naik Lab | New York, NY | February 2023 - June 2024*

- Built and maintained Snakemake workflows for bulk RNA-seq, ATAC-seq, single-cell RNA-seq, CITE-seq, and spatial transcriptomics.
- Integrated scRNA-seq and spatial transcriptomics data to examine cellular composition and spatial organization in human and mouse skin.
- Built a QIIME 2 pipeline for mouse 16S rRNA sequencing studies and analyzed microbial diversity and taxonomic composition in R, including skin wound-healing and gut microbiome datasets.
- Analyzed ChIP-seq data to identify enriched genomic regions and compare signals across experimental conditions.
- Managed code, data, and documentation for laboratory sequencing projects and external collaborations.
- Prepared figures and analysis results for manuscripts, presentations, and grant applications.
- Advised postdoctoral fellows and graduate students on study design, sequencing analysis, and interpretation.
- Assisted the principal investigator with peer review of manuscripts, focusing on bioinformatics methods and interpretation.
- Co-first authored a *Science Immunology* paper on spatial transcriptomics in psoriasis and contributed analyses to additional publications.

## Bioinformatics Analyst | NYU Langone Health

*Applied Bioinformatics Laboratories - Naik Lab | New York, NY | July 2019 - January 2023*

- Analyzed bulk RNA-seq, ATAC-seq, scRNA-seq, CITE-seq, and spatial transcriptomics data for collaborative research projects.
- Developed differential-expression and statistical-analysis workflows and prepared heatmaps, UMAPs, spatial maps, and other scientific figures.
- Contributed computational analysis and figures to peer-reviewed publications.

## Research Assistant | Boston University Medical Campus

*Computational Biomedicine | Boston, MA | February 2019 - May 2019*

- Analyzed single-cell RNA-seq data from adult mouse tracheal epithelium using Bayesian hierarchical modeling for cell-type classification.

## Hub Analyst / Research Assistant | Boston University

*Bioinformatics Hub | Boston, MA | June 2018 - February 2019*

- Analyzed bulk RNA-seq data from post-mortem brain samples from studies of CTE, Huntington's disease, and Parkinson's disease.
- Developed Snakemake workflows for differential-expression analysis and prepared PCA, heatmap, and pathway-analysis figures.

## EDUCATION

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**Master of Science in Bioinformatics**, Boston University, Boston, MA | 2019

**Bachelor of Science in Bioinformatics**, Panjab University, Chandigarh, India | 2017

## SELECTED PUBLICATIONS

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- 1 **Castillo, R.L., Sidhu, I., et al.** Spatial transcriptomics stratifies psoriatic disease severity by emergent cellular ecosystems. *Science Immunology* 8, eabq7991 (2023). **Co-first author.**
- 2 Konieczny, P. et al. Interleukin-17 governs hypoxic adaptation of injured epithelium. *Science* (2021).
- 3 Subudhi, I. et al. Metabolic coordination between skin epithelium and type 17 immunity sustains chronic skin inflammation. *Immunity* 57, 1665-1680.e7 (2024).
- 4 Wasko, R. et al. Langerhans cells are essential components of the angiogenic niche during murine skin repair. *Developmental Cell* 57, 2699-2713.e5 (2022).
- 5 Mascharak, S. et al. Inhibiting mechanotransduction prevents scarring and yields regeneration in a large animal model. *Science Translational Medicine* 17 (2025).

Complete publication list: ORCID 0000-0002-1470-6421

## LANGUAGES

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English | Hindi | Punjabi